10.1. Measure an angle in radian

Definition and results

The trigonometric circle is the center circle O and radius 1, oriented in the inverse direction
A radian corresponds to an arc of length 1 on this circle.
value 2π radians, half a turn π. On a radius circle r, a length of arc is $r\theta$ for one positive angle θ expressed in radians.

Status and justification. Definitions and proportionality between centre angle and arc length. Conversion is 180 $^\circ = \pi$ radians; multiplies the degrees by $\frac{\pi}{180}$.

Example solved. $60^\circ = \frac{\pi}{3}$; $\frac{3\pi}{2} = 270^\circ$. An angle of $\frac{\pi}{3}$ on a circle of radius 6 cm gives an arc of 2 π Inch.

Point of vigilance

The length of an arc is not that of the rope. In trigonometric calculation, check the mode degree or radian of tool.

Immediate application. Convert 45 $^\circ$, 120° ; give the radius arc 2 m and angle $\frac{3\pi}{4}$.

10.2 Winding and image of a real

Definition and results

To each real x corresponds an M point of the circle obtained by a length oriented course x since I(1;0). Two reals have the same image if their difference is an integer multiple of 2 π.
A negative number indicates a route in the clockwise direction.

Status and justification. Adding or removing an integer number of turns keeps the point of arrival. Conversely, returning to the same point requires an integer number of turns.

Example solved. $\frac{13\pi}{3} = 4\pi + \frac{\pi}{3}$ has the same image as $\frac{\pi}{3}$. $-\frac{\pi}{2}$ has the same image as $\frac{3\pi}{2}$: point $(0; -1)$.

Point of vigilance

Add π gives the opposite point, not the same point.

Immediate application. Place images of $\frac{7\pi}{2}$ and $-\frac{5\pi}{3}$ Bringing them back to a turn.

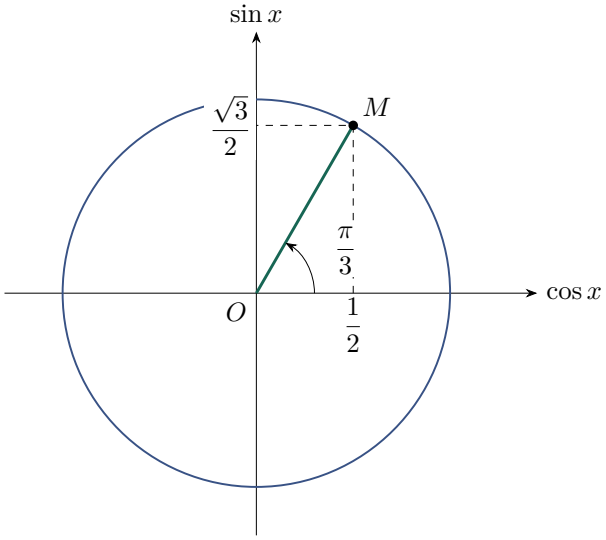
10.3 Sine, Cosine and Remarkable Values

Definition and results

If M is the image of x, $M = (\cos x; \sin x)$ Therefore $-1 \leq \cos x, \sin x \leq 1$ and $\cos^2 x + \sin^2 x = 1$.
For $x = 0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$, the cosine are 1 $, \frac{\overline{3}}{2}, \frac{\overline{2}}{2}, \frac{1}{2}$, 0 and sine 0 $, \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, 1$.

Status and justification. À $\frac{\pi}{4}$, the two coordinates are equal and positive ; Pythagoras gives $2a^2 = 1$, therefore $a = \frac{\sqrt{2}}{2}$. An equilateral triangle on side 1, cut by its height, has half-base $\frac{1}{2}$ and height $\frac{\sqrt{3}}{2}$: $\cos(\frac{\pi}{3}) = \frac{1}{2}$, $\sin(\frac{\pi}{3}) = \frac{\sqrt{3}}{2}$. Additional angle gives $\frac{\pi}{6}$.

Example solved. The angle point $\frac{\pi}{3}$ is $(\frac{1}{2}; \frac{\sqrt{3}}{2})$ The scalar product of two unit vectors making this angle is worth its cosine, so $\frac{1}{2}$. The geometry chapters thus connect to the circle.



For angle $\frac{\pi}{3}$, the projections of M give directly its cosine and sine.
placed outside the lines of the drawing.

Point of vigilance

The cosine is the abscisse, the sine is the ordinate. The identity gives an absolute value: the quadrant determines the sign.

Immediate application. If $\cos x = \frac{3}{5}$ and image of x is in the fourth quadrant, giving $\sin x$.

10.4 Associated angles and symmetries

Definition and results

The symmetries of the circle give $\cos(-x) = \cos x$, $\sin(-x) = -\sin x$; $\cos(\pi - x) = -\cos x$, $\sin(\pi - x) = \sin x$; $\cos(\pi + x) = -\cos x$, $\sin(\pi + x) = -\sin x$. Add 2 $k\pi$, k integer, does not change no sine or cosine.

Status and justification. Each equality is read on the coordinates of the symmetrical points: axis horizontal for $-x$, vertical axis for $\pi - x$, centre O for $\pi + x$. This is not a catalogue Without justification.

Example solved. $\cos(\frac{5\pi}{6}) = -\frac{\sqrt{3}}{2}$, $\sin(\frac{5\pi}{6}) = \frac{1}{2}$ by symmetry of $\frac{\pi}{6}$. $\cos(-\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$, $\sin(-\frac{\pi}{4}) = -\frac{\sqrt{2}}{2}$.

Point of vigilance

Two angles of the same cosine do not necessarily have the same sine.
read the signs.

Immediate application. Calculate the two coordinates of the images of $\frac{4\pi}{3}$ and $-\frac{\pi}{4}$.

10.5 Approximate pi and reinvest

Definition and results

A regular polygon inscribed n sideways in the half-perimeter unit circle $n\sin(\frac{\pi}{n})$; a polygone circumscribed to half-perimeter $n\tan(\frac{\pi}{n})$, with $\tan\theta = \frac{\sin\theta}{\cos}\theta$ Here. Archimedes method doubles the number of sides and calculates the new lengths by Pythagoras, without knowing pi In advance.

Status and justification. The marked perimeter is lower than that of the circle and circumscribes it top: geometric frame allowed. If it is the rope on one side, place $h = \sqrt{1 - \frac{c^2}{4}}$; the rope of half-angle is $\frac{c}{\sqrt{2(1+h)}}$. This formula comes from Pythagoras and avoids subtraction of real close.

Example solved. For the listed hexagon, c=1 and half-perimeter 3. $\sqrt{3} \approx$ 3,4641. By doubling the sides, the bounds is tightened. The programme provided starts from n=6 and c=1.

Point of vigilance

Directly using a sine function with already known pi would check a calculation, but does Would not build an independent pi approximation.

Immediate application. For n=6 and c=1, calculate the new string after doubling.